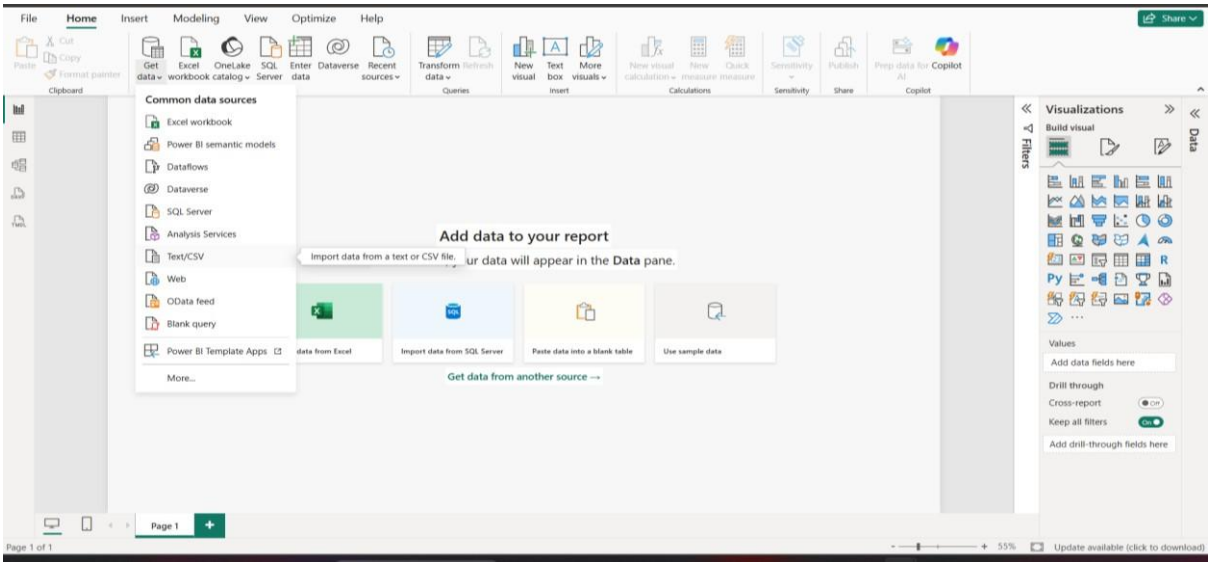
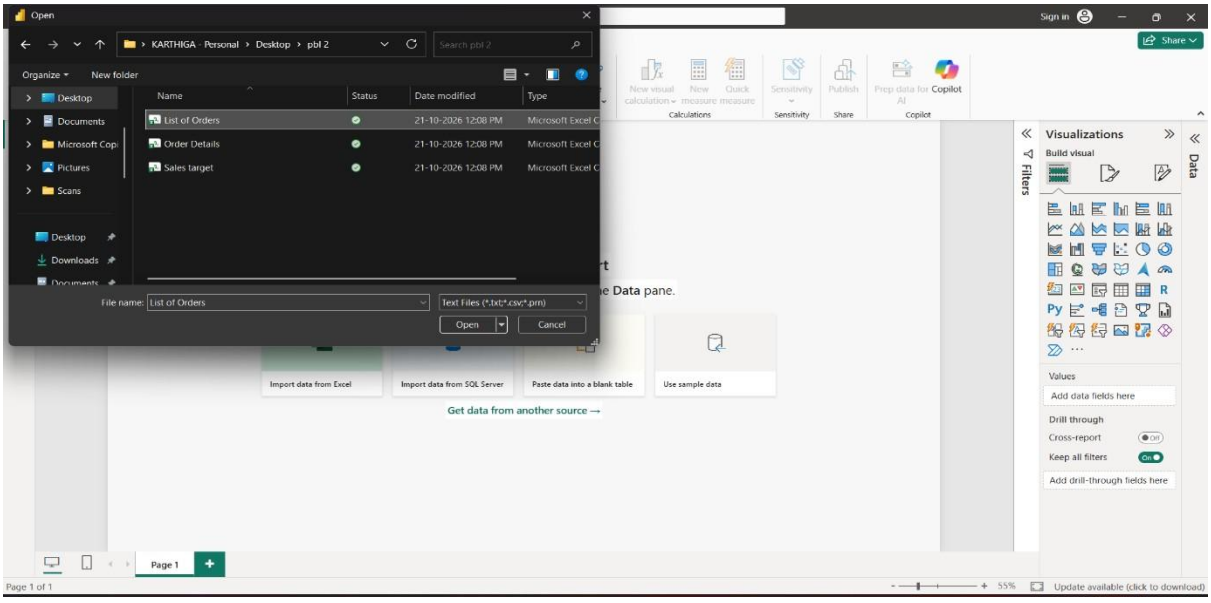
Power BI Assignment 2 – DAX & Data Visualization

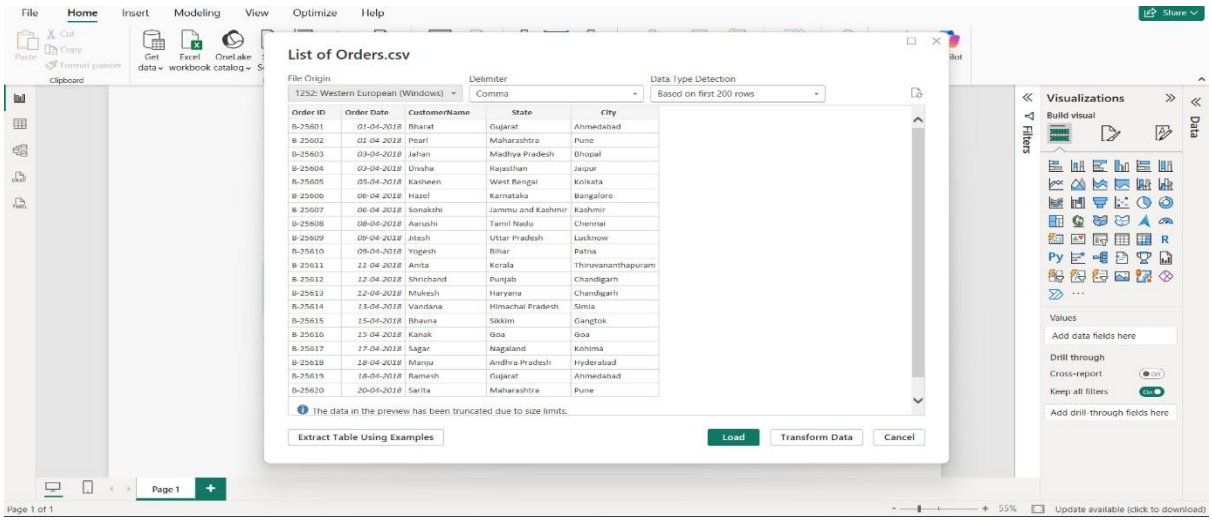
Here , the three E- Commerce Sales Analysis datasets are given CSV format .

- ★ **List of Orders.csv**
- ★ **Order Details.csv**
- ★ **Sales target.csv**

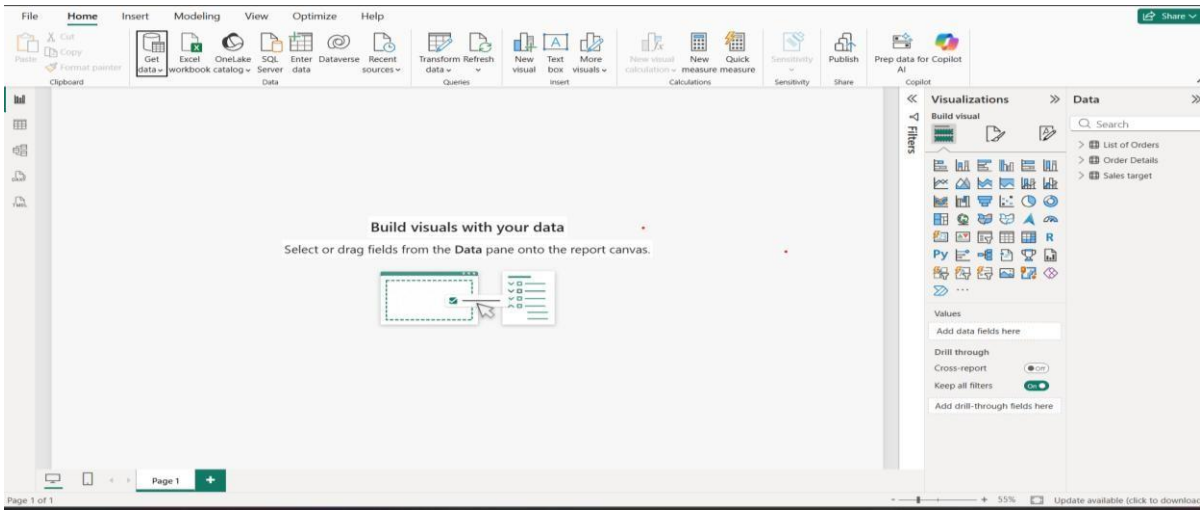
These files are imported into the Power BI and into the Power Query Editor.

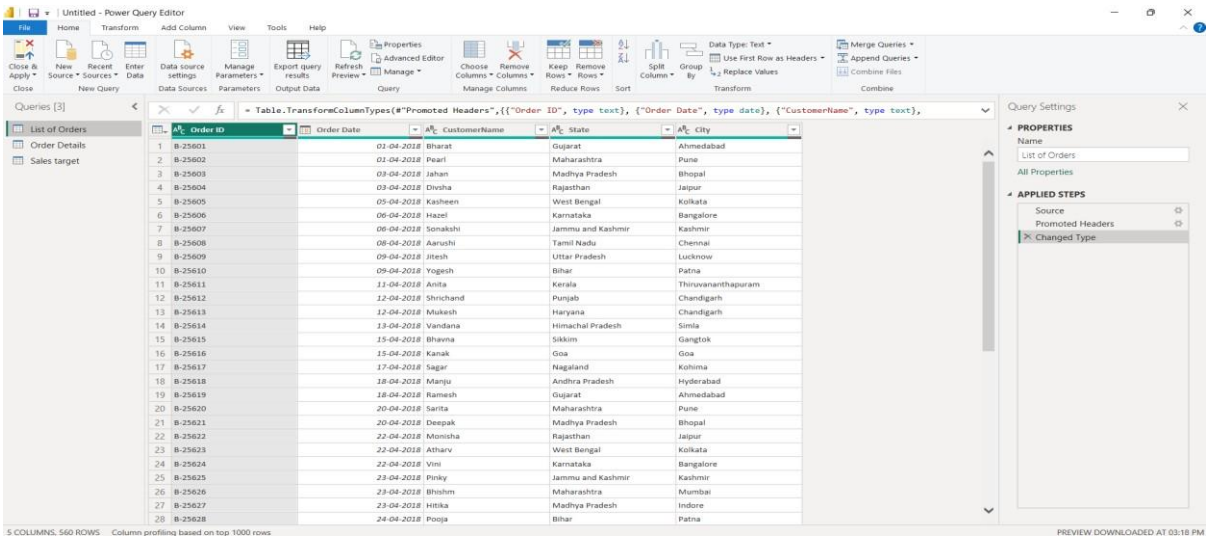




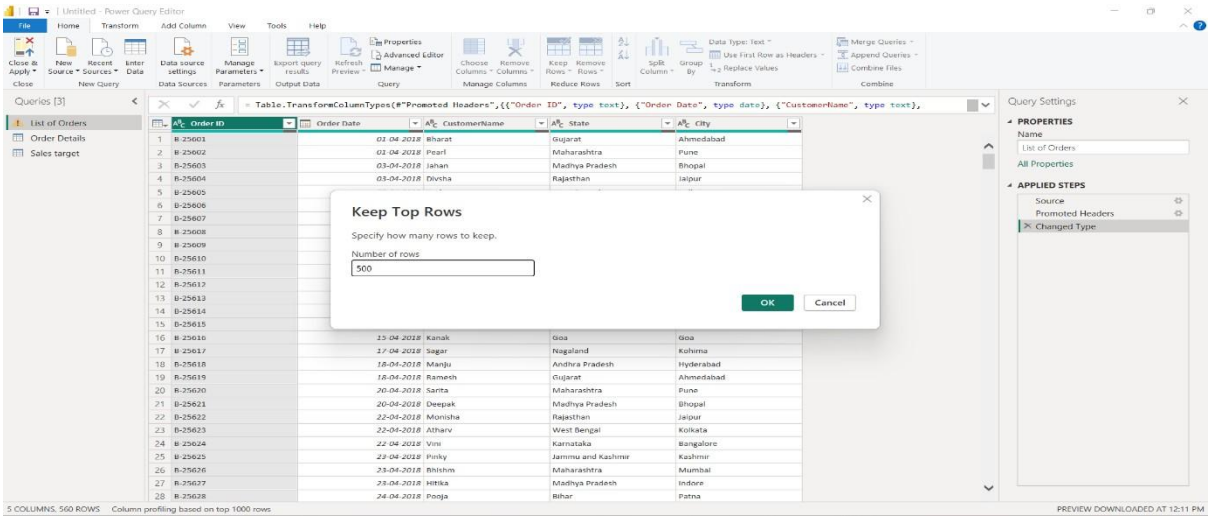


The same procedure is repeated for the remaining 2 CSV files and are imported into the PowerBI and Power Query Editor.

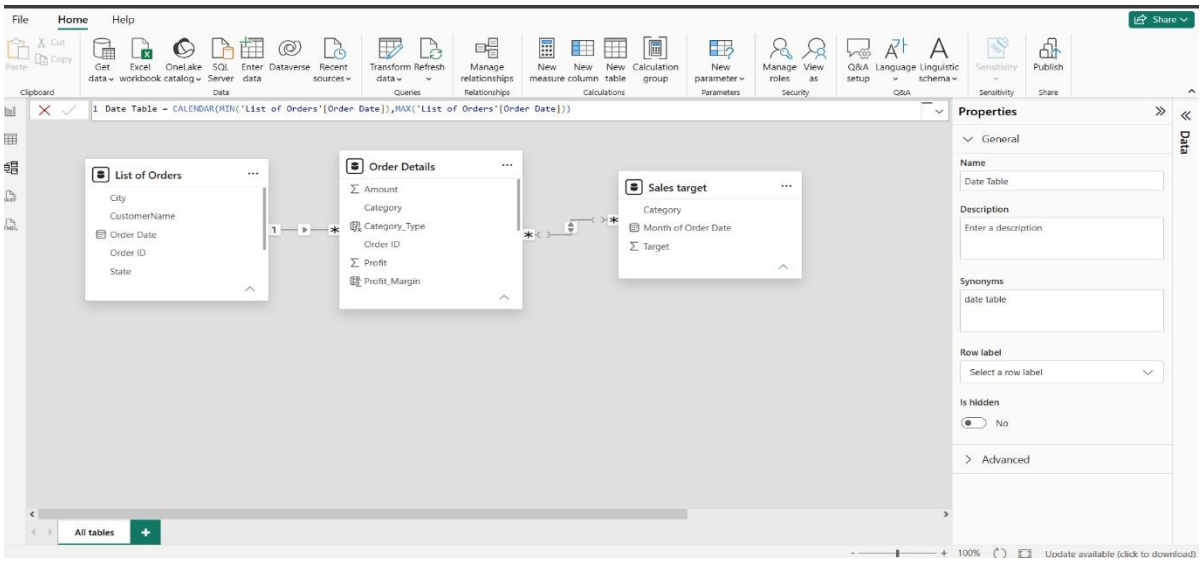




In the LIST OF ORDERS table , there are 560 rows totally with 60 NULL valued rows at the bottom . So, we can remove those NULL valued rows and keep top 500 rows for further Analysis.



DATA MODELING



The Relationship between the given 3 datasets are established.

List Of Orders (Order Id) ----- Order Details (Order Id)

Cardinality – Many to One Cross Filter Directions – Single

Order Details (Category) ----- Sales Target (Category)

Cardinality – Many to Many Cross Filter Directions – Both

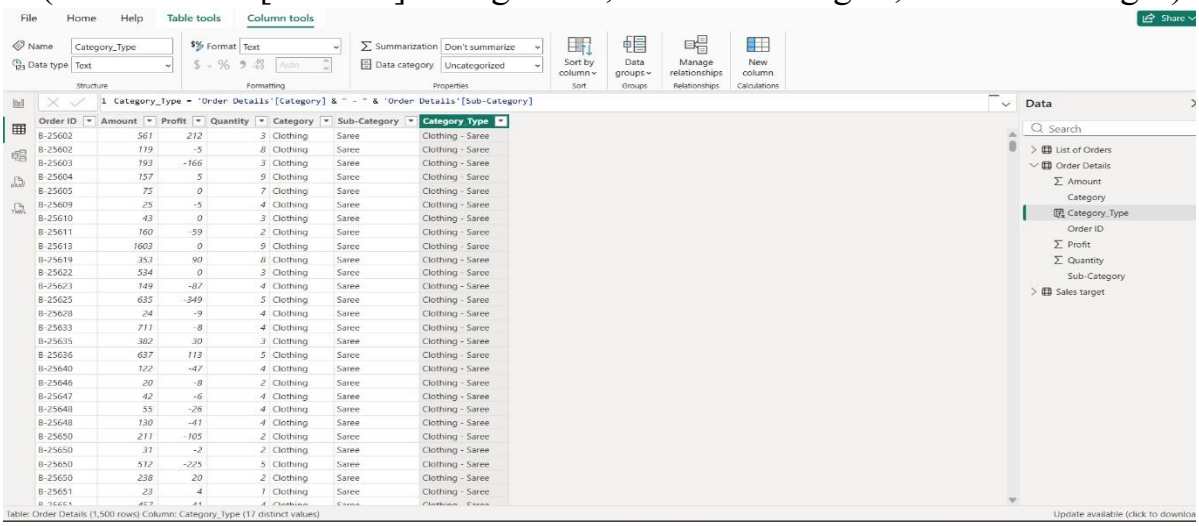
CALCULATED COLUMNS

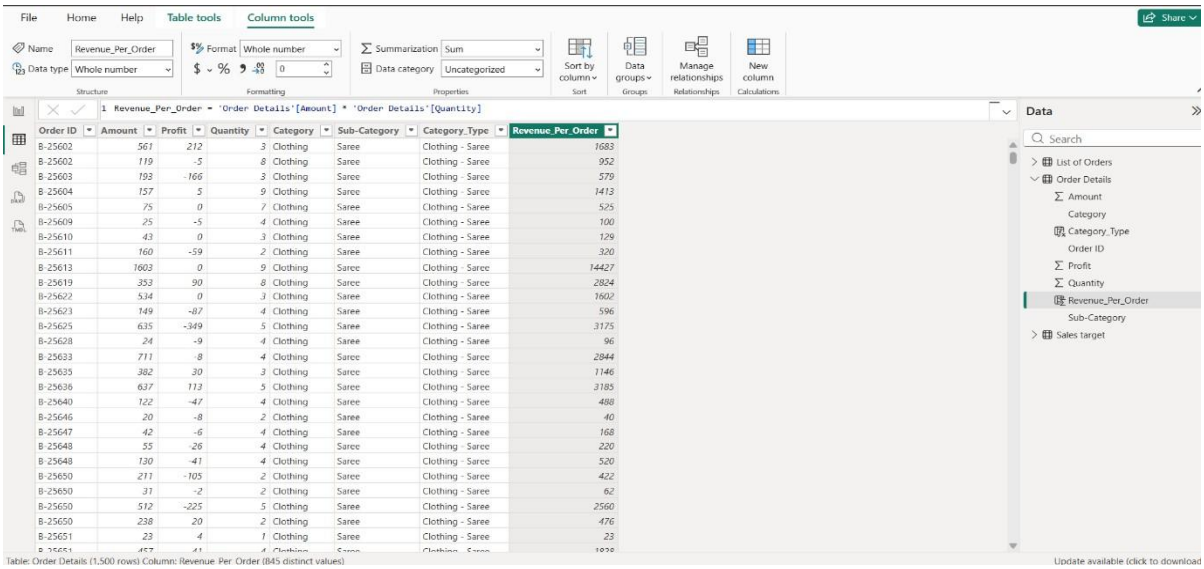
In the ORDER DETAILS table , the 3 columns are created and are named as CATEGORY_TYPE , REVENUE_PER_ORDER , SALES_CATEGORY .

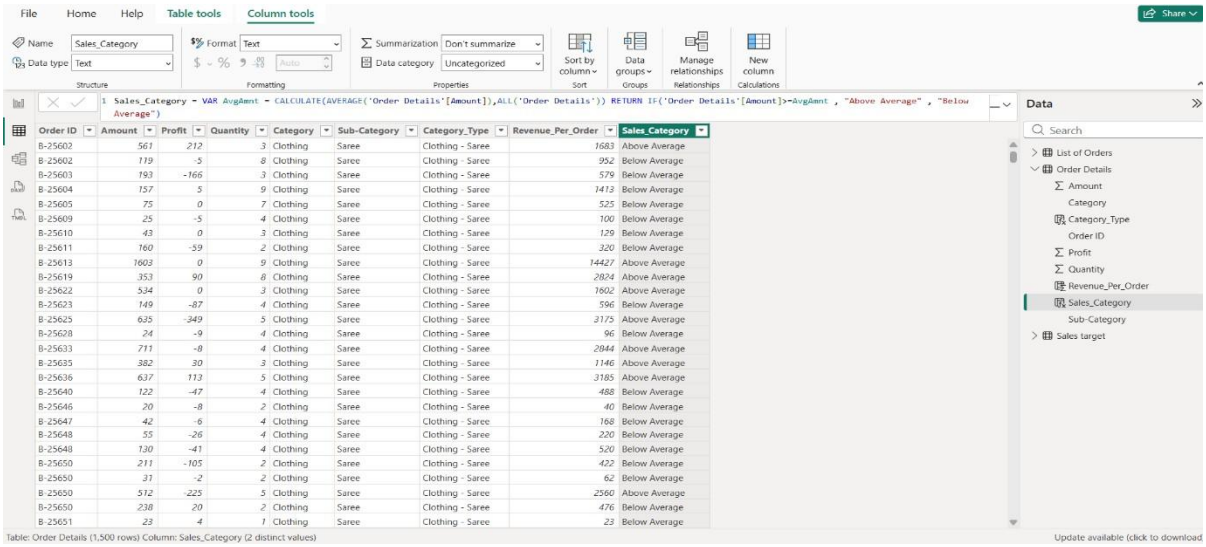
CATEGORY _TYPE = 'Order Details'[Category] & " - " & 'Order Details'[SubCategory]

REVENUE_PER_ORDER = 'Order Details'[Amount] * 'Order Details'[Quantity]

SALES_CATEGORY = VAR AvgAmnt = CALCULATE(AVERAGE('Order Details'[Amount]),ALL('Order Details')) RETURN IF('OrderDetails'[Amount]>=AvgAmnt , "Above Average" , "Below Average")







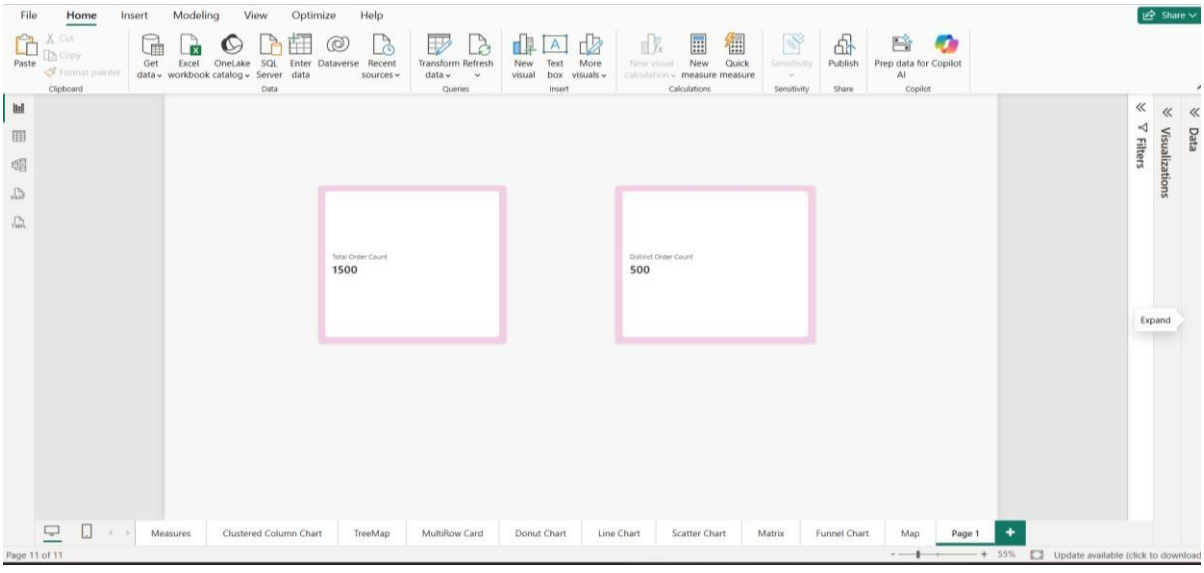
CALCULATED MEASURES

1. CALCULATE ORDER COUNT

In the ORDER DETAILS Table , the COUNTROWS () and DISTINCTCOUNT () functions are used to find the total order count and the distinct order count.

TOTAL ORDER COUNT = COUNTROWS('Order Details')

DISTINCT ORDER COUNT = DISTINCTCOUNT('Order Details'[Order ID])

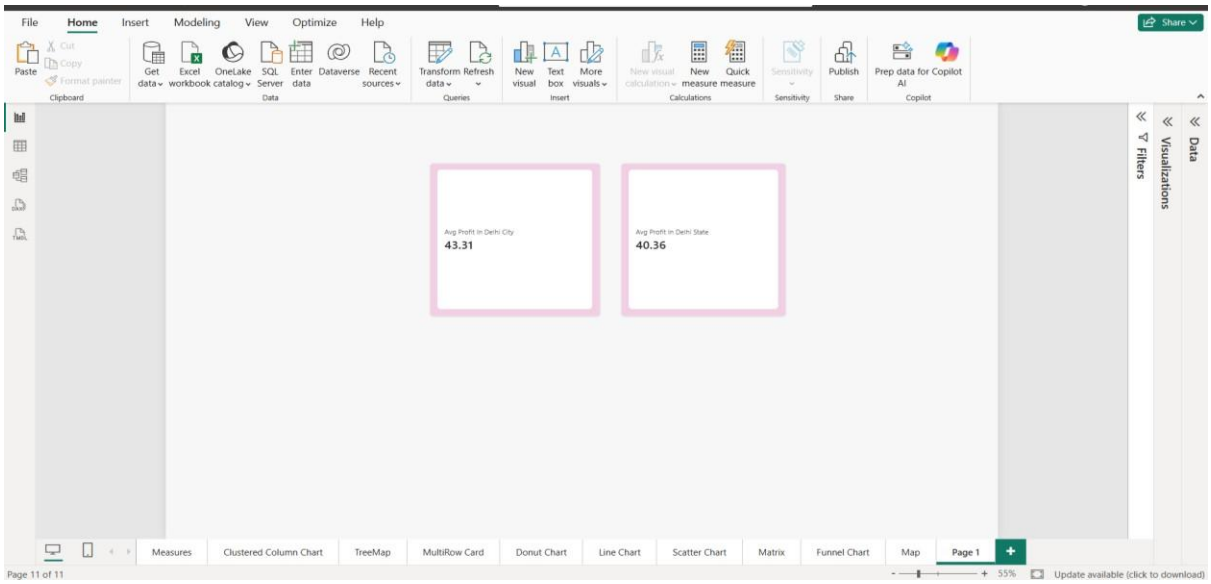


2. CALCULATE AVERAGE PROFIT IN DELHI

In the LIST OF ORDERS Table , DELHI is presented in both CITY and STATE columns. So, 2 measures are calculated.

Avg Profit In Delhi City = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[City] = "Delhi")

Avg Profit In Delhi State = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[State] = "Delhi")



3. CALCULATE YEAR TO DATE (YTD) SALES

Here, the New table named DATE TABLE is created using the NEW TABLE option. This table contains 5 columns named DATE, YEAR, MONTH, MONTH_NAME, MONTH_YEAR.

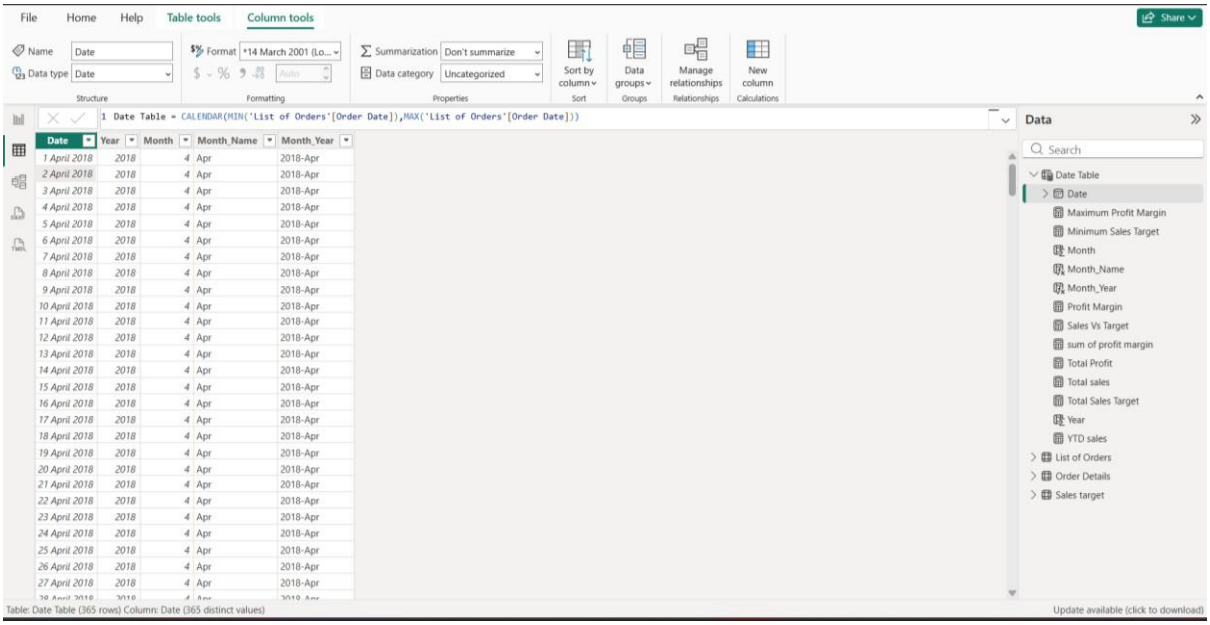
DATE TABLE = CALENDAR(MIN('List of Orders'[Order Date]),MAX('List of Orders'[Order Date]))

MONTH = MONTH('Date Table'[Date])

YEAR = YEAR('Date Table'[Date])

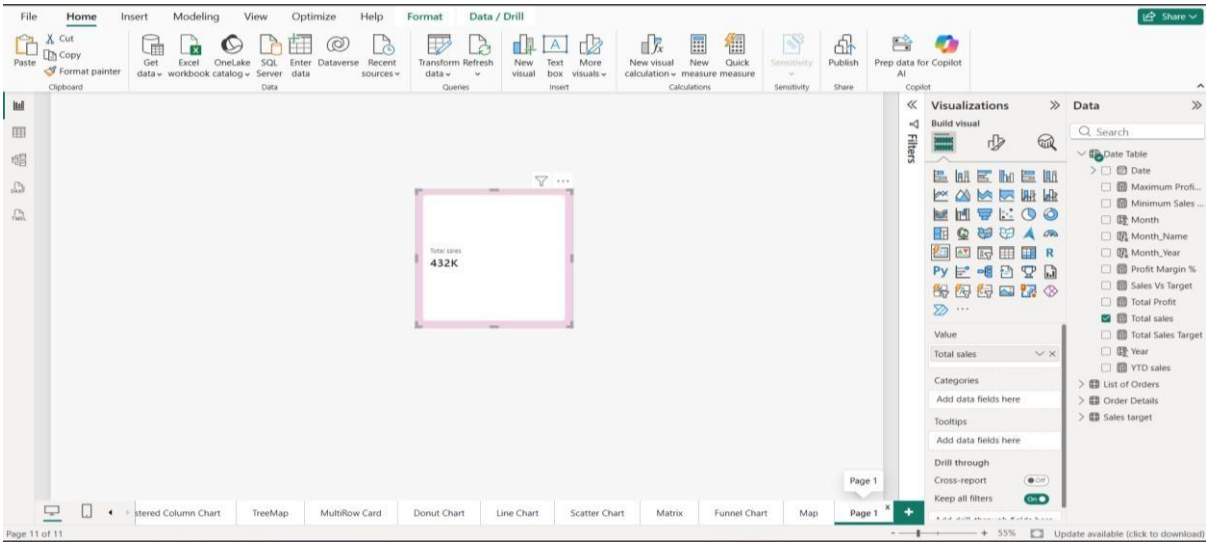
MONTH_NAME = FORMAT('Date Table'[Date], "MMM")

MONTH_YEAR = FORMAT('Date Table'[Date], "YYYY-MMM")



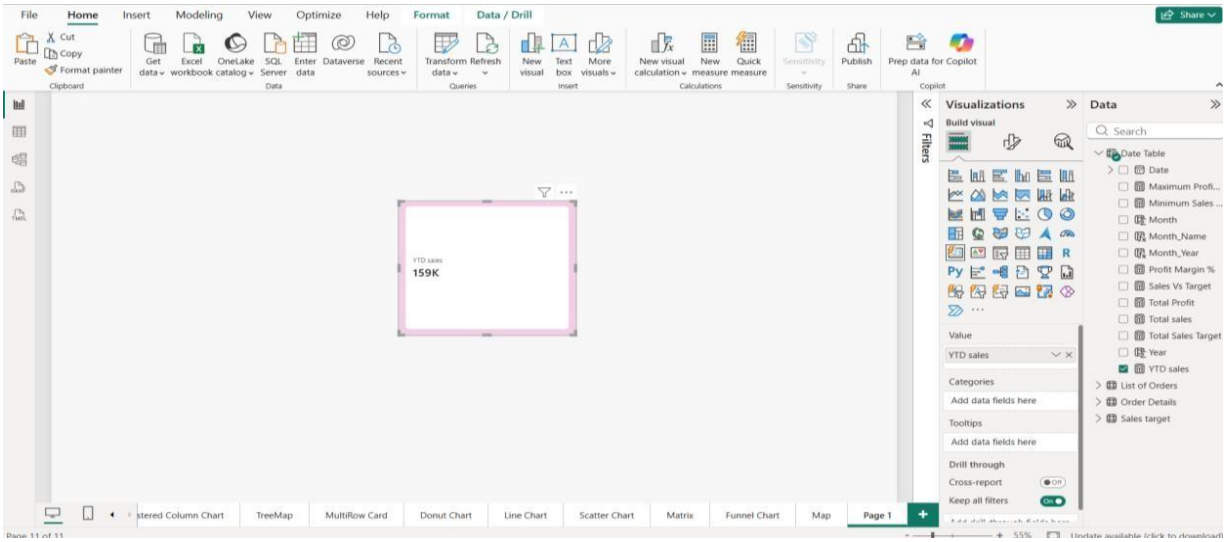
Now, TOTAL SALES is created as a NEW MEASURE.

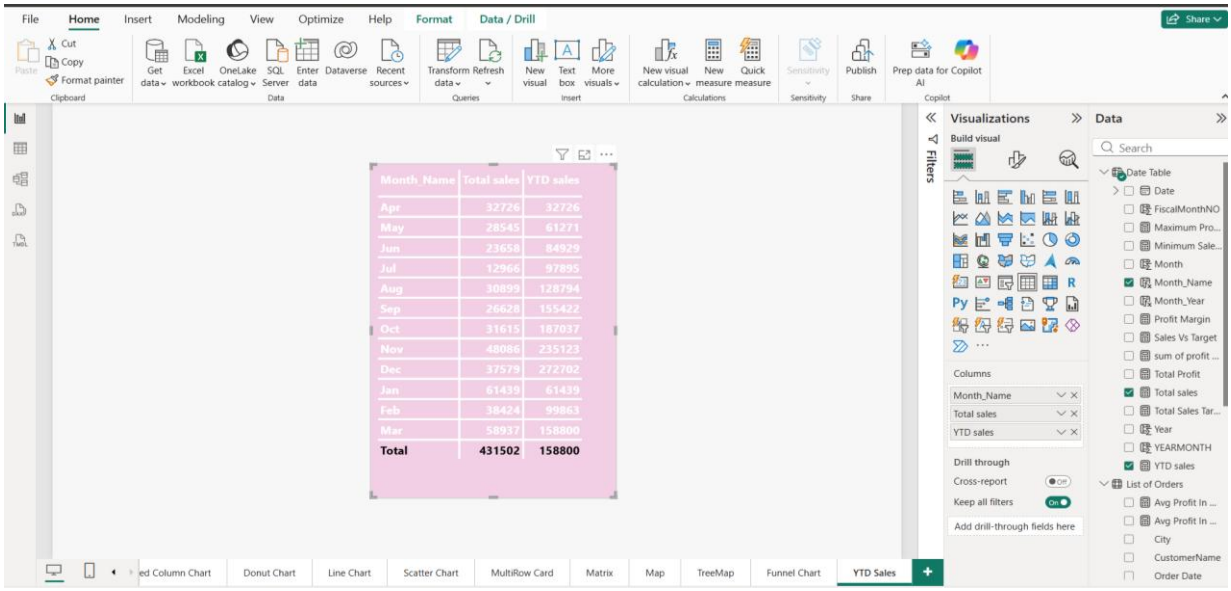
TOTAL SALES = SUM('Order Details'[Amount])



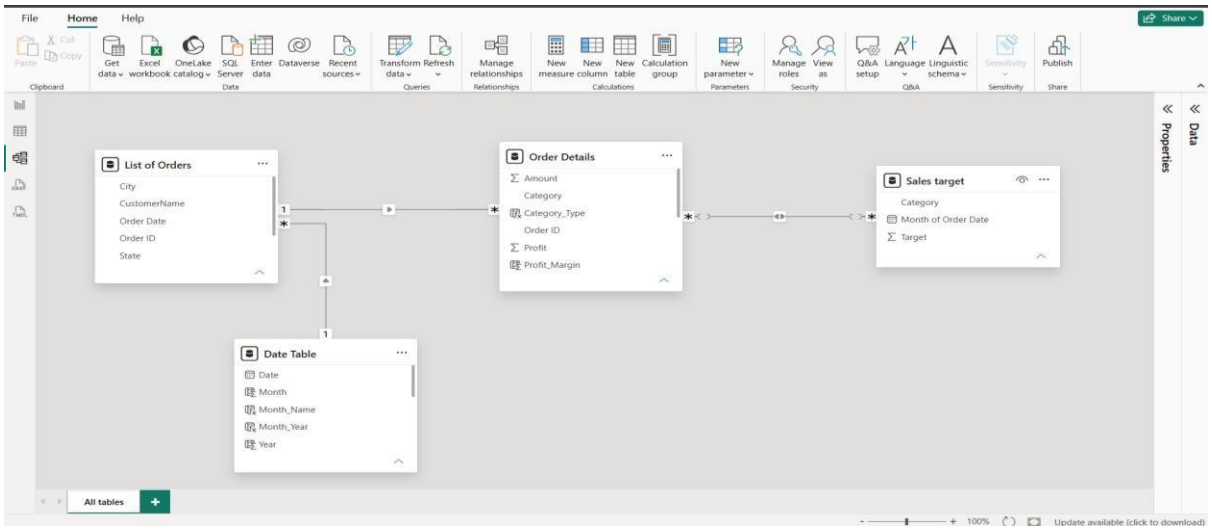
Now , YTD SALES measure is calculated using the formula.

YTD SALES = TOTALYTD([Total sales],'Date Table'[Date])





The Relationship is created between all the 4 tables ., LIST OF ORDERS, ORDER DETAILS, SALES TARGET AND DATE TABLE.



List Of Orders (Order Date) ----- Date Table (Date)

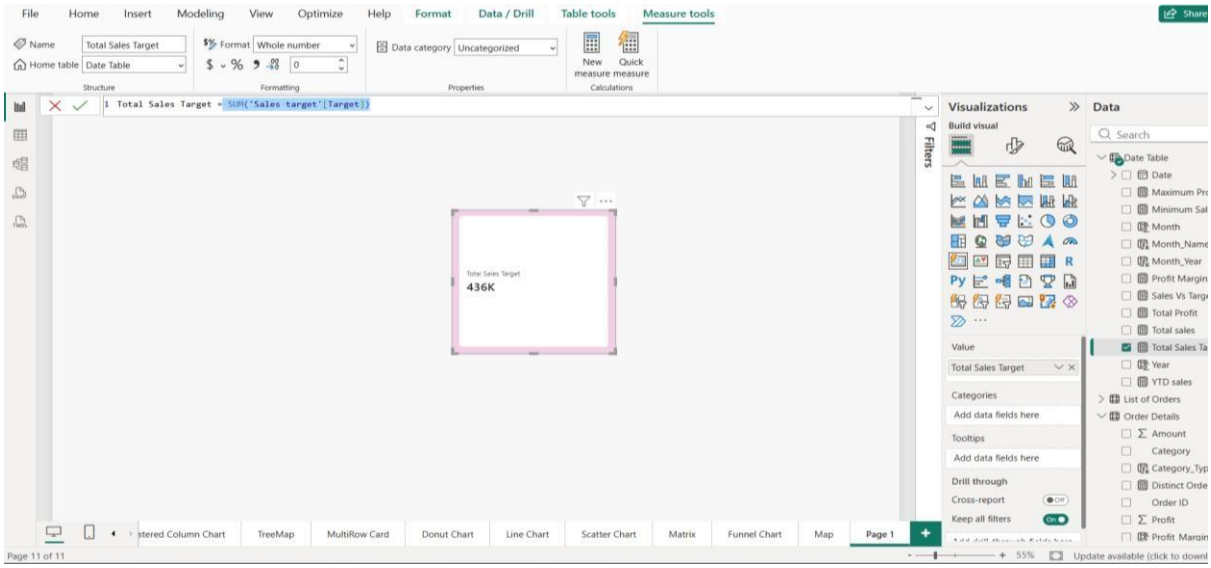
Cardinality = Many to One Cross Filter Directions = Single

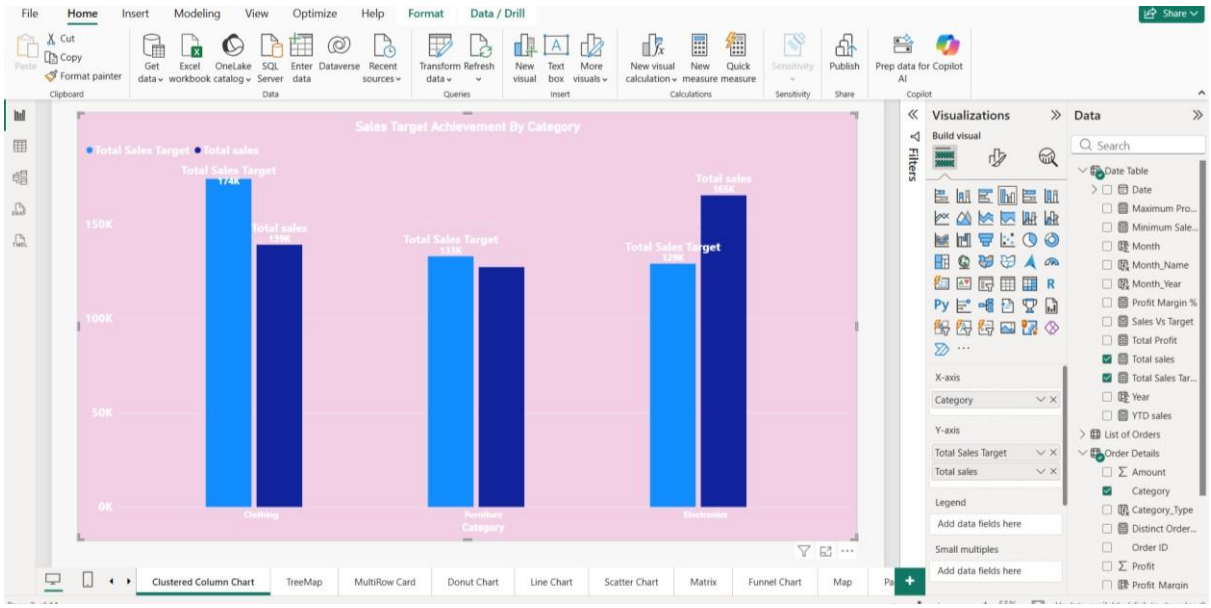
DATA VISUALIZATION

1. **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.

Here . new measure TOTAL SALES TARGET is created.

TOTAL SALES TARGET = SUM('Sales target'[Target])





2. **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

For this chart , the new column PROFIT_MARGIN is created in the ORDER DETAILS table.

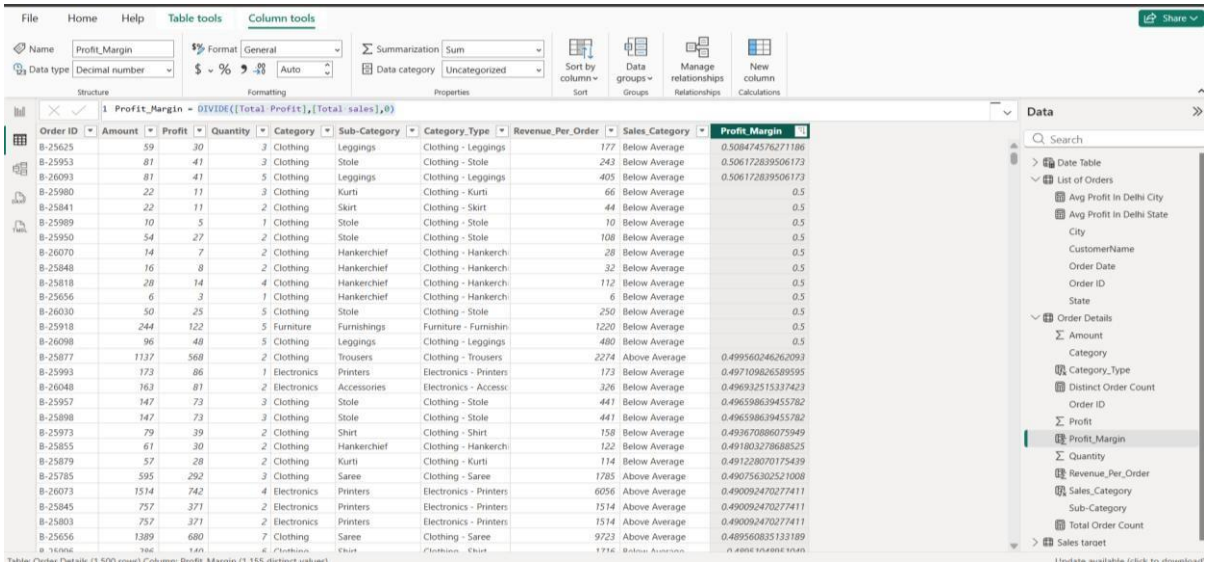
PROFIT_MARGIN = DIVIDE([Total Profit],[Total sales],0)

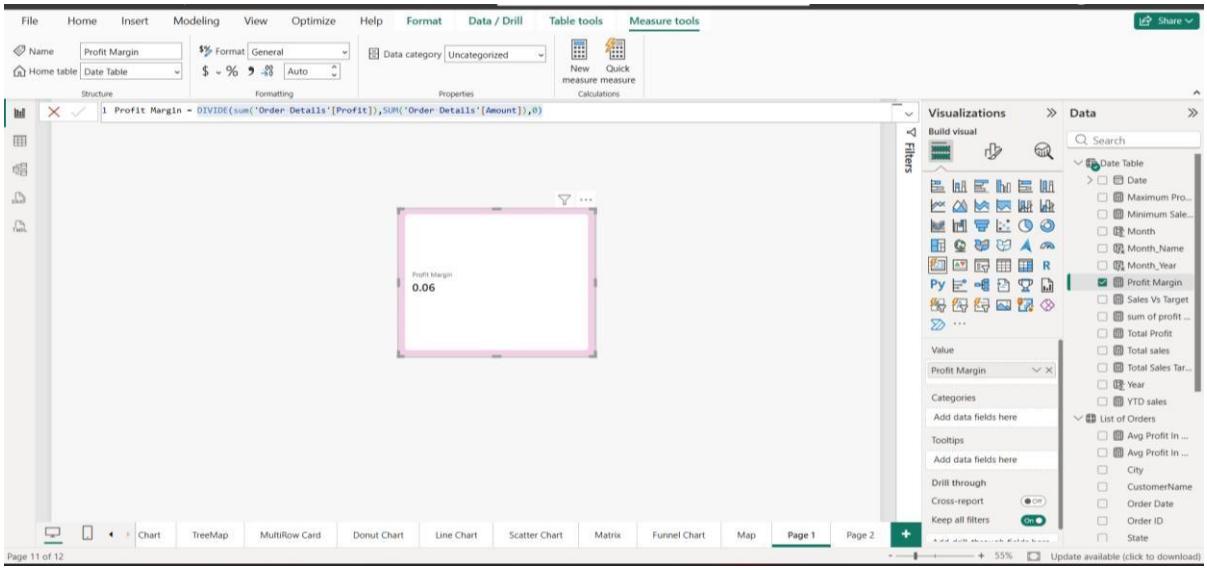
New 3 measures are created.

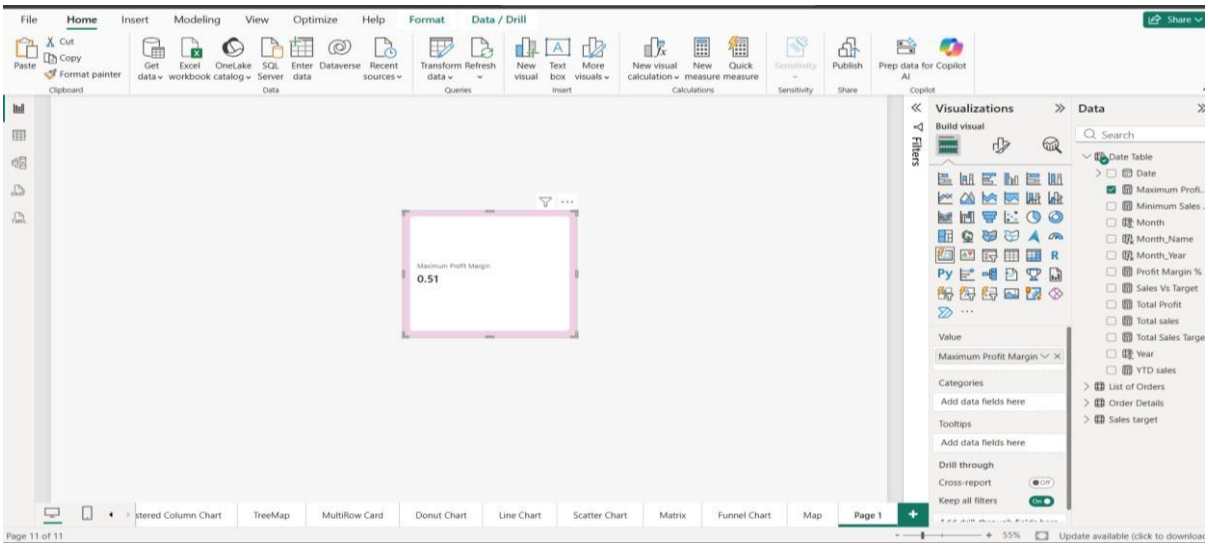
PROFIT MARGIN = DIVIDE (SUM ('Order Details'[Profit]), SUM ('Order Details'[Amount]), 0)

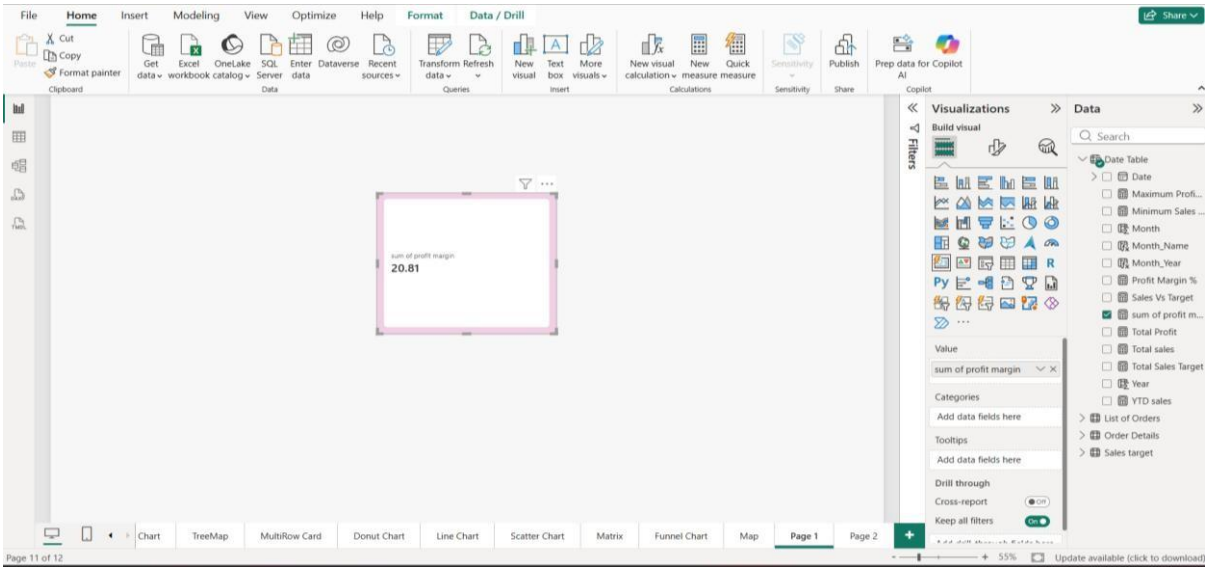
MAXIMUM PROFIT MARGIN = MAX('Order Details'[Profit_Margin])

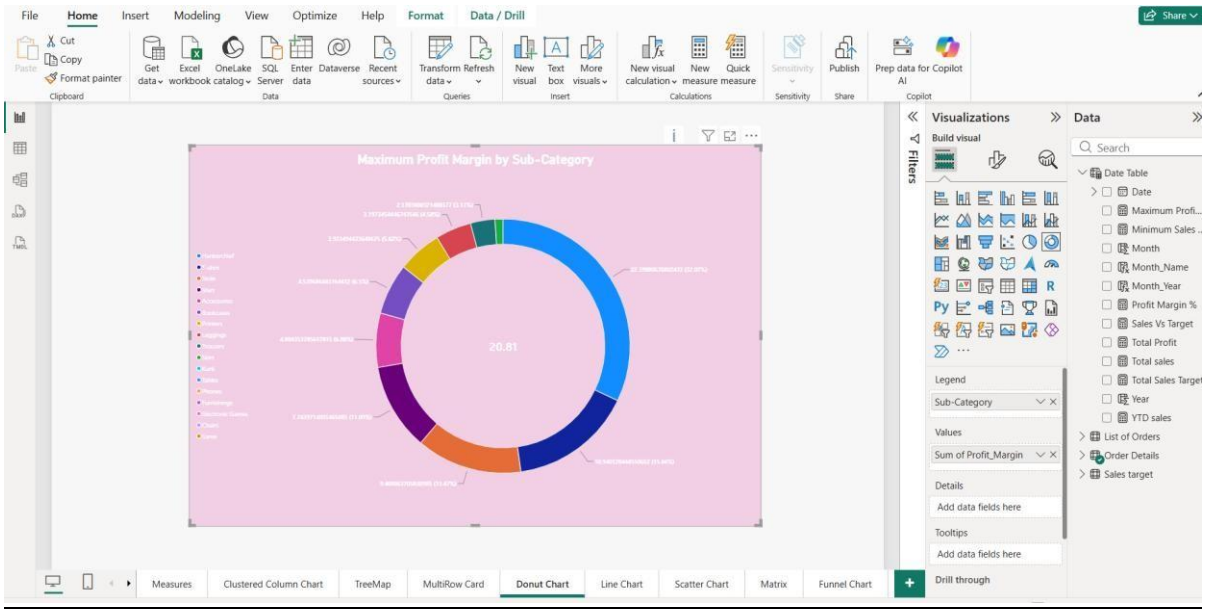
SUM OF PROFIT MARGIN = SUM('Order Details'[Profit_Margin])



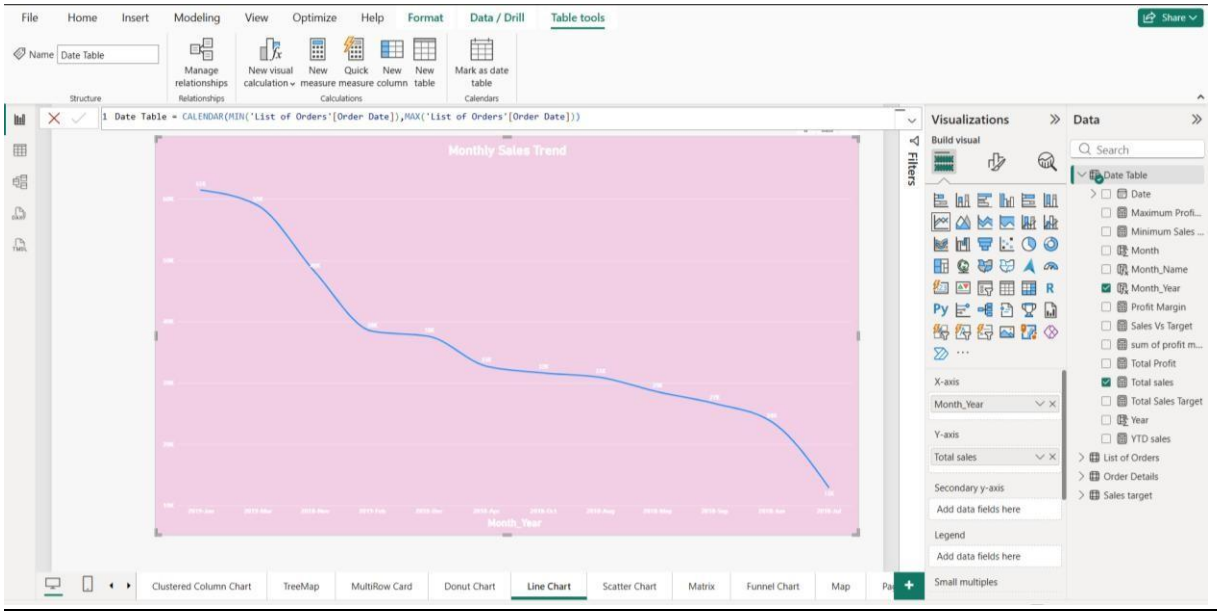

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools
Profit Margin = DIVIDE(sum('Order Details'[Profit]),SUM('Order Details'[Amount]),0)
Profit Margin
0.06
Visualizations
Data
Page 11 of 12


File Home Insert Modeling View Optimize Help Format Data / Drill
Maximum Profit Margin
0.51
Visualizations
Data
Page 11 of 11

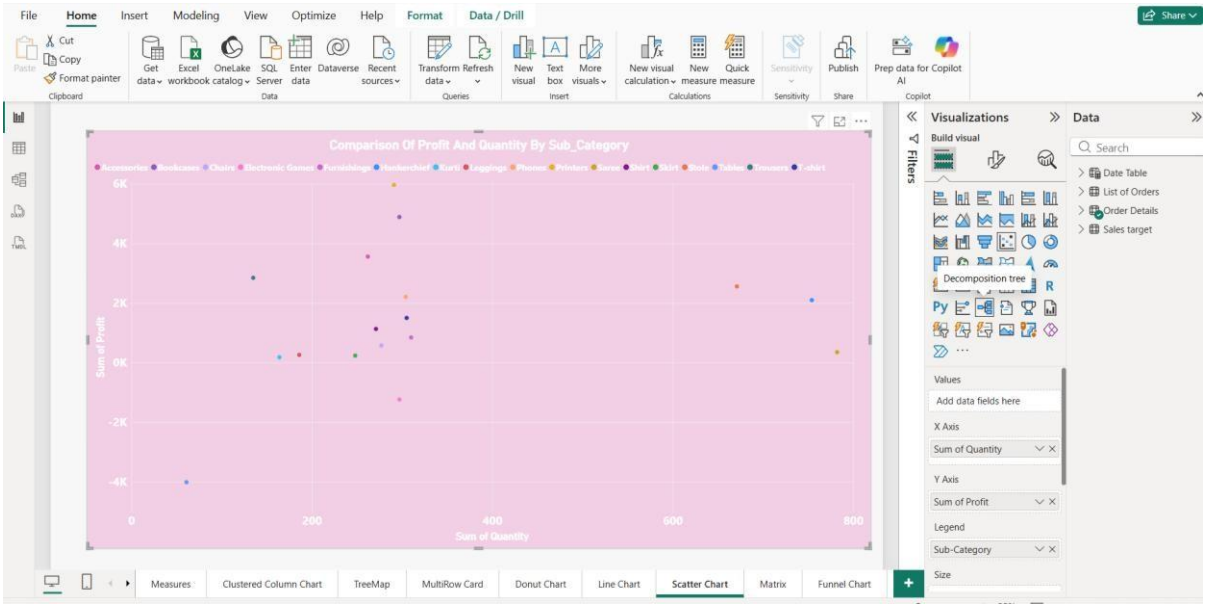

File Home Insert Modeling View Optimize Help Format Data / Drill
sum of profit margin
20.81
Visualizations
Data
Page 11 of 12



3. **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart

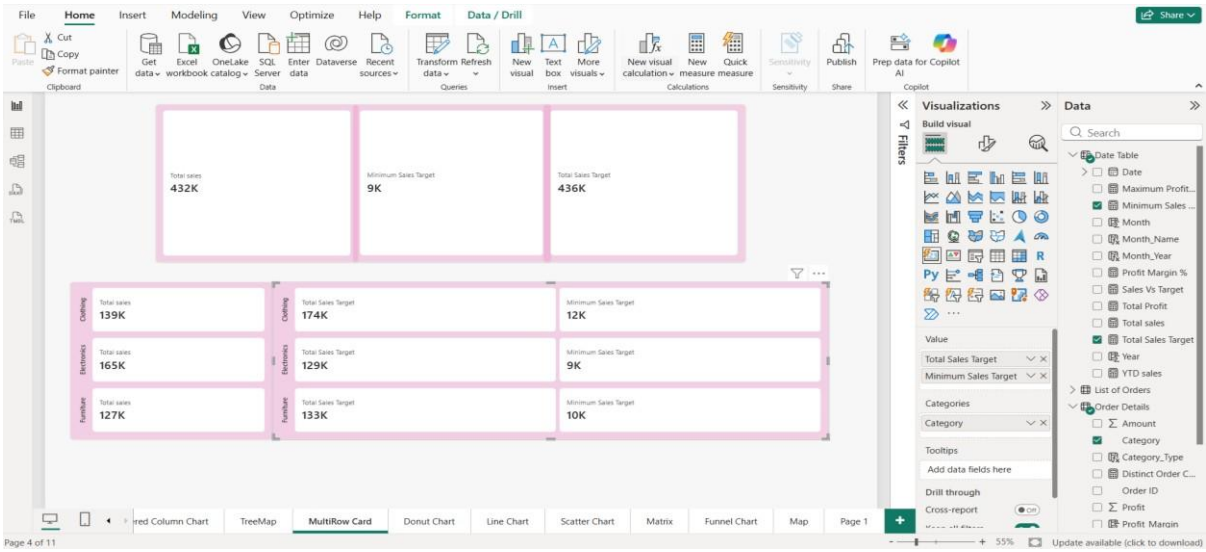


4. **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different subcategories using a scatter chart.



5. **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

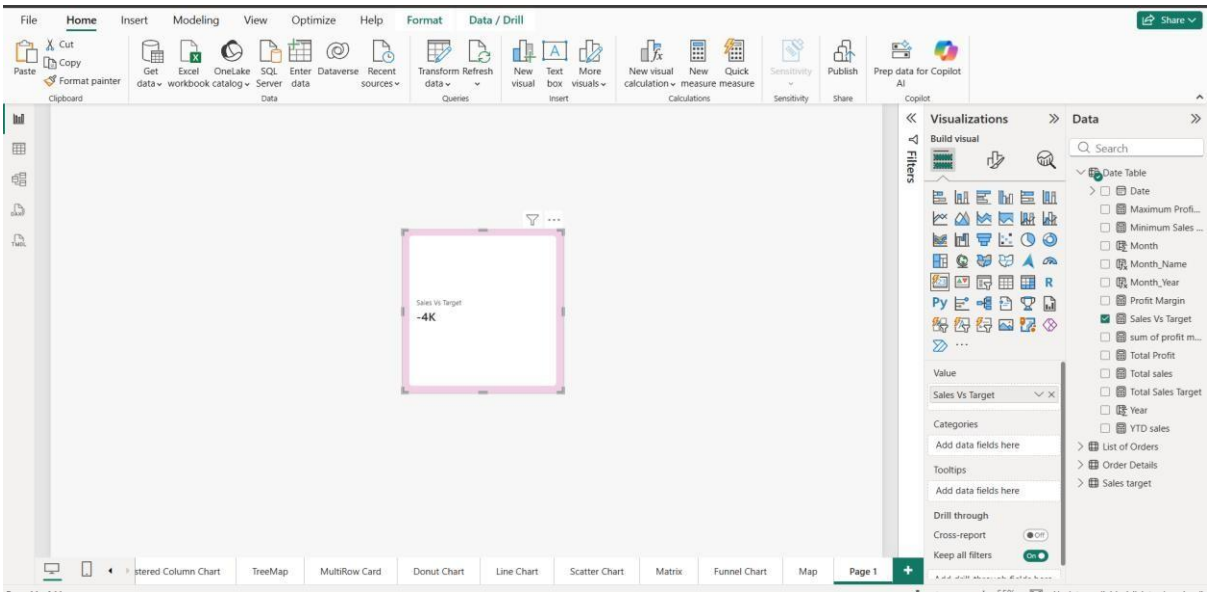
Total Sales Target = SUM('Sales target'[Target])
Minimum Sales Target = MIN('Sales target'[Target])

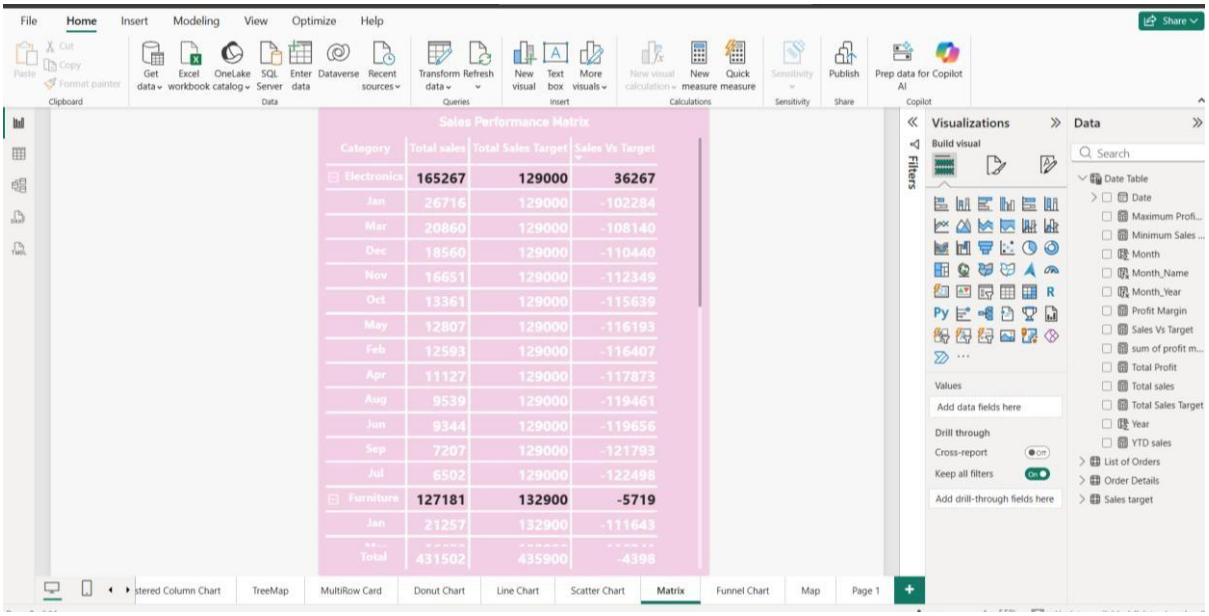


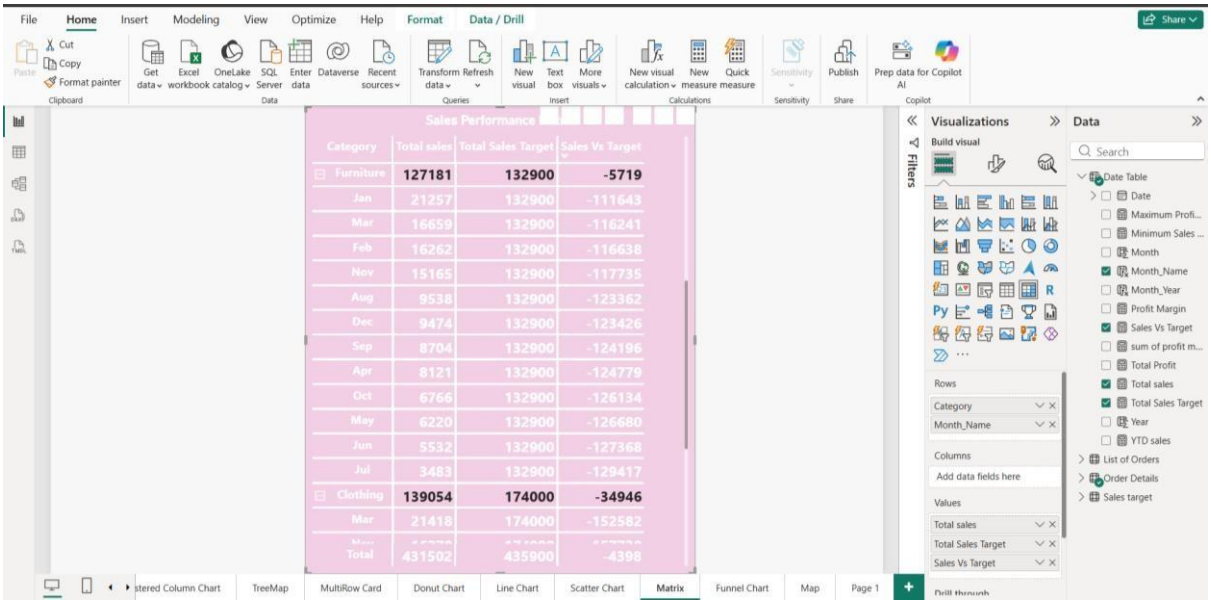
6. **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

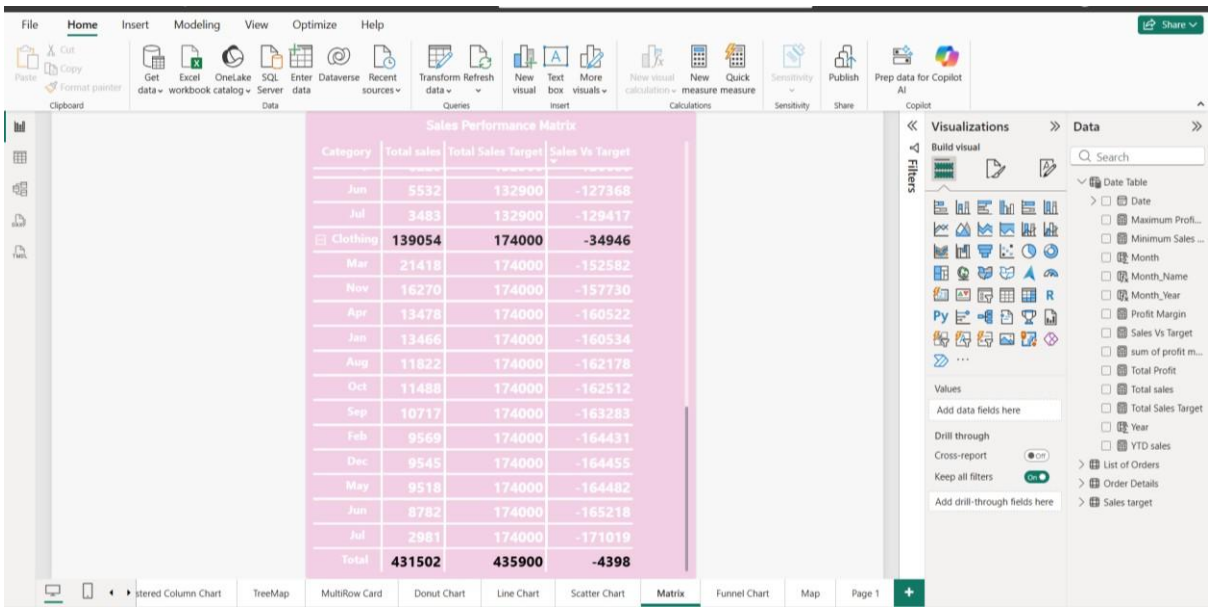
The new measure is created.

SALES VS TARGET = [Total sales] - [Total Sales Target]

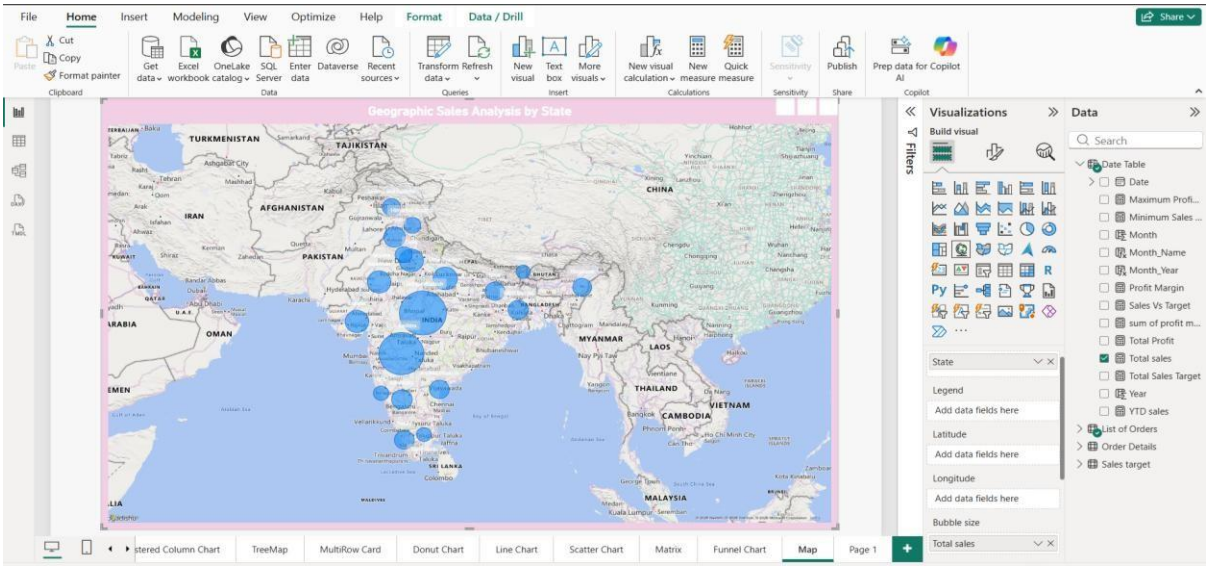




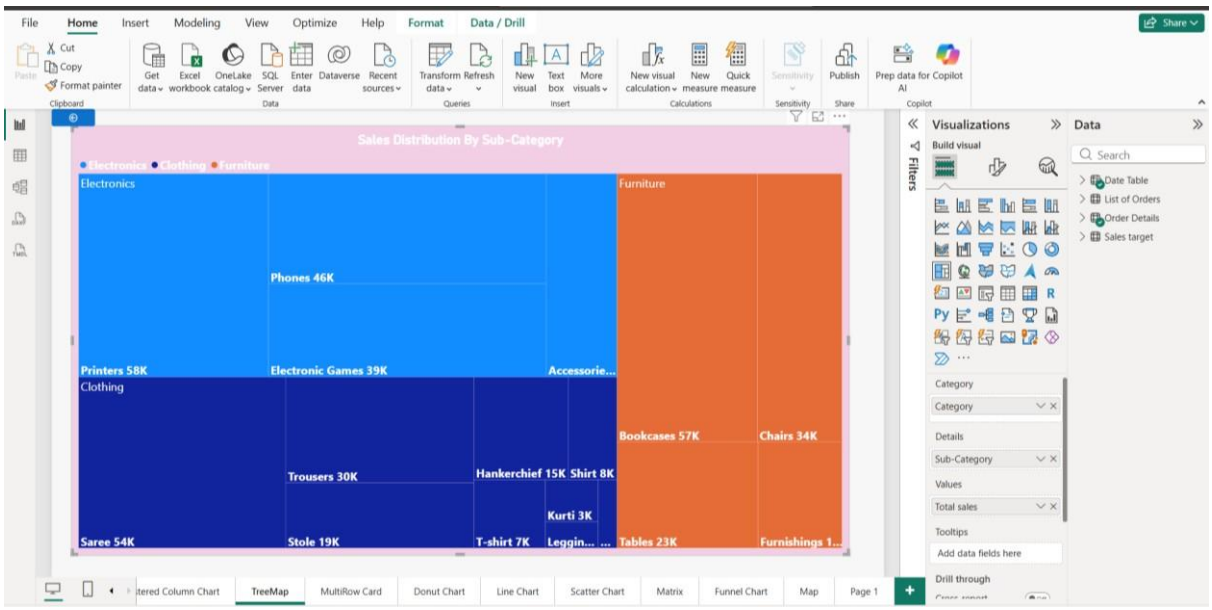




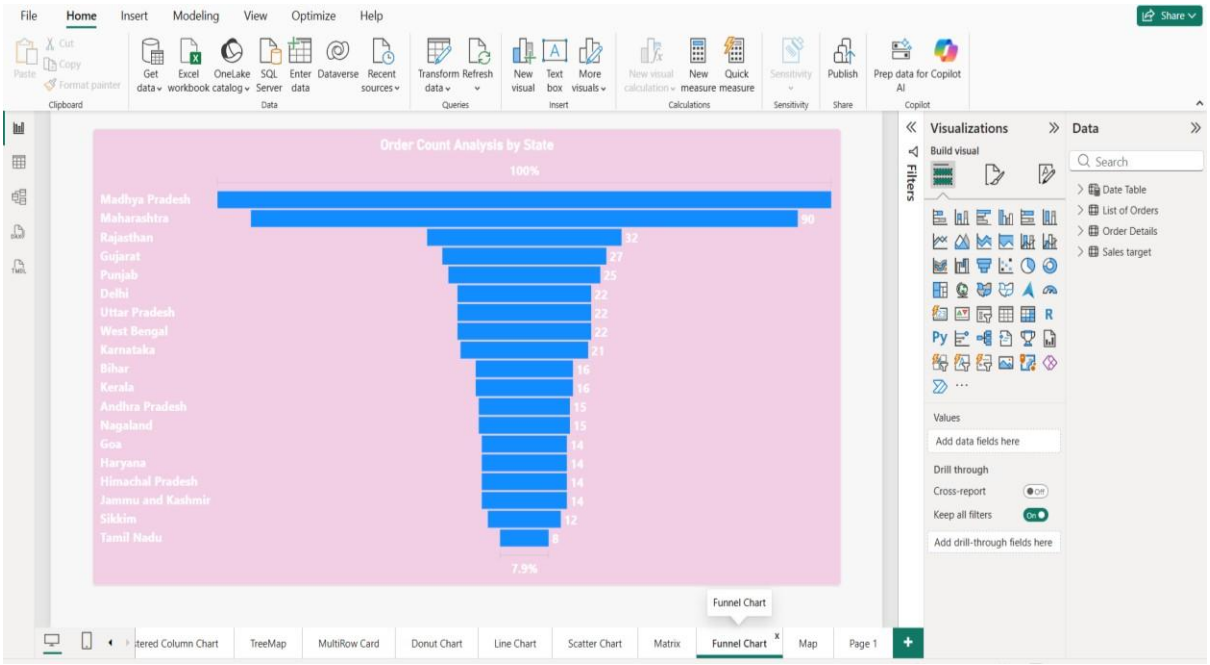
7. **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.



8. **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a Treemap.



9. **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.



Finally the measures found out are ,

